JGi JAIN DEEMED-TO-BE UNIVERSITY | FACULTY OF ENGINEERING AND TECHNOLOGY

Department of Computer Science & Engineering

Global Campus, Jakkasandra Post, KanakapuraTaluk, Ramanagara District, Pin Code: 562 112

MULTI-LABEL EMOTION DETECTION IN TEXT USING RoBERTa WITH IMBALANCE-AWARE OPTIMIZATION

- PRESENTED BY :

  NAMALA LAKSHMI PRASANNA(23BTRCL224)

  I LEELA VAISHNAVI(23BTRCL103)

  LIKITHA H(23BTRCL051)

  KEERTHI KM(23BTRCL106)

  JEEVAN M(23BTRCL194)

  JAI BALAJI (23BTRCL104)

- GUIDED BY : DR P RENUKADEVI

  Assistant Professor / CSE- AIML

CONTENTS

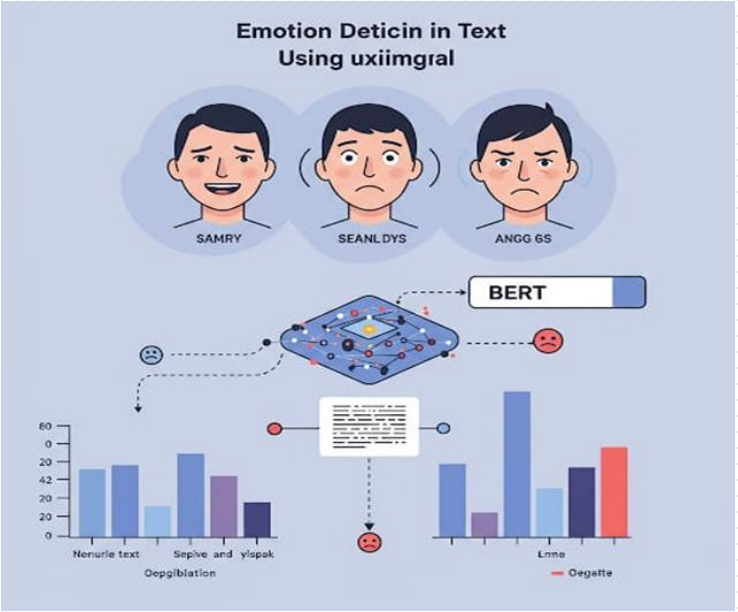
- *ABSTRACT*
- *INTRODUCTION*
- *PROBLEM STATEMENT*
- *EXISTING SYSTEM*
- *BLOCK DIAGRAM*
- *LITERATURE SURVEY*
- *PROPOSED SYSTEM*
- *BLOCK DIAGRAM*
- *PROPOSED SYSTEM OVERVIEW*
- *CONCLUSION*
- *REFERENCE*

ABSTRACT

- Emotion detection in text has evolved into a **multi-label classification problem**, where a single sentence can express multiple emotions simultaneously.
- This work proposes a **transformer-based framework using RoBERTa-large** for fine-grained emotion detection on the **GoEmotions dataset (121K samples, 27 emotions + neutral)**.
- To address **class imbalance**, the model uses **Binary Cross-Entropy with class-balanced pos_weight**, improving performance on minority emotions.
- An **adaptive threshold tuning approach** is applied to optimize the **Micro F1-score** instead of using a fixed threshold.
- The model is evaluated using **F1-score, precision, and recall**, demonstrating strong performance and scalability for real-world applications.

INTRODUCTION

- Rapid growth of **social media and online text** has increased the need for emotion detection
- Emotion detection identifies **multiple emotions** such as joy, anger, sadness, etc.
- Real-world text often expresses **multiple emotions simultaneously**, making it a **multi-label classification problem**
- Traditional methods (lexicon-based, SVM, CNN) lack **deep contextual understanding**
- RoBERTa captures deep contextual relationships using self-attention
- However, challenges like **class imbalance** and **threshold selection** still affect performance
- This work focuses on building an **imbalance-aware, transformer-based emotion detection system**



PROBLEM STATEMENT

- Existing systems mainly focus on **single-label or basic sentiment classification**, missing fine-grained emotions

- Real-world text often contains **multiple overlapping emotions**, requiring **multi-label classification**

- The **GoEmotions dataset is highly imbalanced**, where some emotions are frequent and others are rare

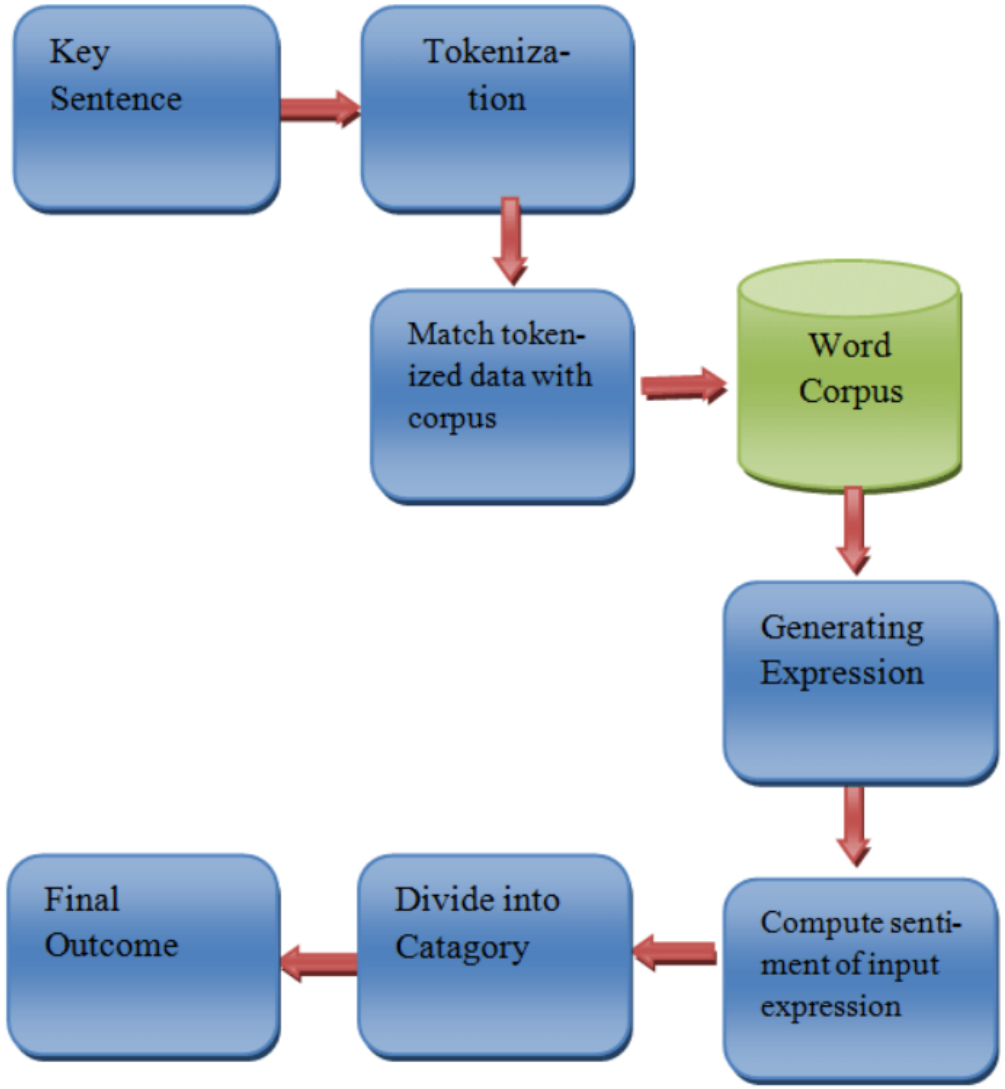
- Traditional and basic deep learning models fail to handle **context, imbalance, and label dependencies effectively**

- Using a fixed threshold (e.g., 0.5) leads to **suboptimal prediction performance**

EXISTING SYSTEM

- Traditional emotion detection systems rely on rule-based approaches or shallow machine learning models like SVM and Naïve Bayes, which lack contextual understanding.
- Many existing models fail to capture the nuances of human emotions due to limited contextual awareness.
- Existing systems often struggle with multilingual text and informal language (e.g., social media posts).
- Lack of real-time usability and user-friendly interfaces for non-technical users.
- Limited scalability and computational efficiency in handling large datasets.
- Not suitable for **multi-label classification**

BLOCK DIAGRAM



LITERATURE SURVEY

STUDY	METHODOLOGY	DATASET USED	ACCURACY	EVALUATION METRICS
A. S. A. Mohamed et al. (2023)	SVM, Naïve Bayes, Decision Trees with TF-IDF	Emotion-labeled text corpus	50 %to 60%	Accuracy, Precision, Recall, F1-score
A. Aldayel et al. (2021)	BERT + CNN hybrid model	Benchmark emotion datasets	60% to 65%	Accuracy, F1-score
B. Y. B. Yogesh et al. (2024)	Deep learning with attention mechanisms	Large-scale text dataset	70%	Accuracy, F1-score
J. Devlin et al. (2019)	BERT (Transformer-based model)	GLUE & NLP benchmarks	68%	Accuracy, F1, Task-specific metrics

PROPOSED SYSTEM

Developed a **multi-label fine-grained emotion detection framework using RoBERTa-large** to capture deep contextual relationships through transformer-based bidirectional self-attention.

The model is trained on the **GoEmotions dataset (121,000 comments, 27 emotions + neutral)** to improve generalization and robustness.

Unlike traditional BERT-based systems, this approach leverages a **larger pre-trained model (RoBERTa-large)** for enhanced semantic understanding and feature representation.

Adaptive **threshold tuning (0.2 – 0.6 range)** is applied to maximize **Micro F1-score**, instead of using a fixed threshold.

BLOCK DIAGRAM

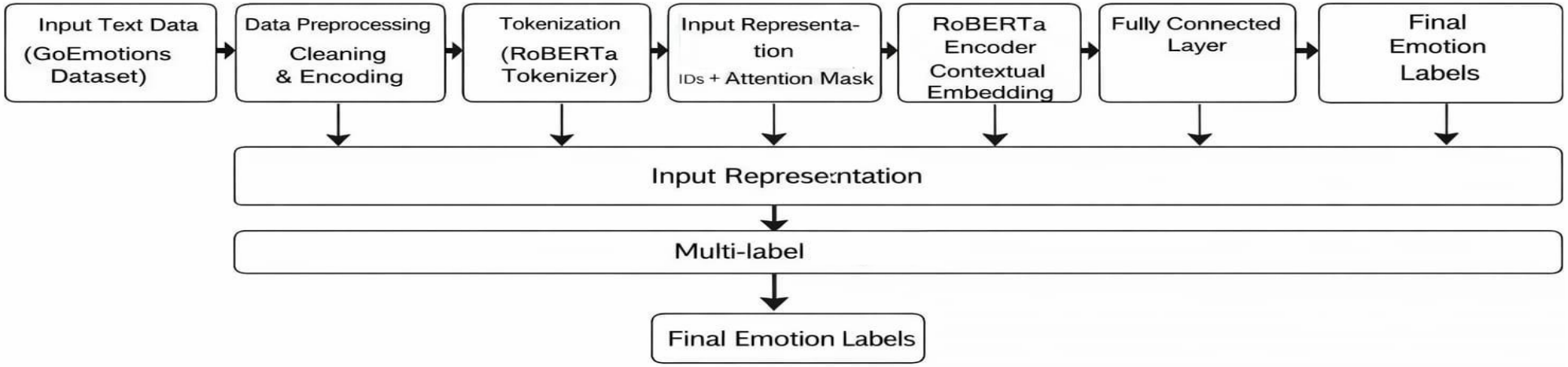


Fig. 1. Proposed multi-label emotion classification architecture using RoBERTa.

PROPOSED SYSTEM OVERVIEW

We propose:

- Transformer-based multi-label classification using **RoBERTa-large**
- Imbalance-aware optimization using **pos_weight (class balancing)**
- **Dynamic threshold tuning** for optimal F1-score
- **End-to-end training using HuggingFace Trainer**
- Scalable architecture for large datasets (121K samples)

Performance metrics:

Micro F1 Score

Macro F1 Score

Precision

Recall

PROCEDURE

1. Data Preprocessing

•Dataset shuffled randomly for better generalization
•Unnecessary columns removed
•Labels converted into **multi-label numerical format**

2. Tokenization & Input Representation

•Text tokenized using **RoBERTa tokenizer**
•Converted into:
•Input IDs
•Attention Masks
•Sequences padded/truncated to **max length = 100**

3. Model Architecture

- **RoBERTa-large transformer model**
- Outputs contextual embeddings
- Final classification layer predicts **multi-label probabilities**

4. Loss Function & Imbalance Handling

- **Binary Cross-Entropy with Logits Loss**
- **pos_weight** applied for handling class imbalance
- Improves learning for minority emotion classes

5. Training Strategy

- Training using **HuggingFace Trainer API**
- Hyperparameters:
  - Learning rate: 2e-5
  - Batch size: 4
  - Epochs: 4
- Mixed precision for faster training

6.Threshold Tuning & Prediction

- Sigmoid applied to logits
- Threshold varied from **0.2 to 0.6**
- Best threshold selected based on **maximum Micro F1-score**
- Final multi-label predictions generated
- Improves balance between **precision and recall**

CONCLUSION

- Proposed a **multi-label emotion detection system** using **RoBERTa-large** for deep contextual understanding
- Handled **class imbalance** using **BCEWithLogitsLoss with pos_weight**
- Applied **adaptive threshold tuning** to improve **Micro F1-score**
- Trained on **full GoEmotions dataset (121K samples)** for better performance
- Achieved results in **F1-score, precision, and recall**
- Demonstrated a **scalable and efficient solution** for real-world emotion detection

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THANK YOU